

logo

Amarm 50219 AgSp1.1 glycine rm1 3  
Msag 18528 AgSp1.2 glycine rm 16  
Aarg AgSp1 FC511 glycine rm1 3  
Mlaby 7048 AgSp1 glycine rm1a 1  
Mhut 1271 AgSp1 glycine rm3 17  
Mlaby 7048 AgSp1 glycine rm1a 6  
Aarg AgSp1 FC511 glycine rm2 7  
Lcorn 18104 AgSp1.1 glycine rm1 39  
Aarg AgSp1 FC511 glycine rm1 15  
Mlaby 7048 AgSp1 glycine rm1a 15  
Varen 6869 AgSp1.1 glycine rm1 2  
Varen 6869 AgSp1.1 glycine rm1 1  
Lcorn 18104 AgSp1.1 glycine rm1 1  
Mlaby 7048 AgSp1 glycine rm1a 2  
Mhut 1271 AgSp1 glycine rm1 1  
Amarm 50219 AgSp1.1 glycine rm1 2  
Mhut 1271 AgSp1 glycine rm1 2  
Amarm 50219 AgSp1.1 glycine rm1 8  
Amarm 50219 AgSp1.1 glycine rm1 1  
Mhut 1271 AgSp1 glycine rm1 3  
Mhut 1271 AgSp1 glycine rm1 5  
Mhut 1271 AgSp1 glycine rm1 7  
Mhut 1271 AgSp1 glycine rm1 4  
Mhut 1271 AgSp1 glycine rm1 6  
Varen 6869 AgSp1.2 glycine rm1 3  
Mlaby 7048 AgSp1 glycine rm2b 21  
Msag 18528 AgSp1.2 glycine rm 1  
Msag 18528 AgSp1.2 glycine rm 115  
Amarm 50219 AgSp1.1 glycine rm2 89  
Amarm 50219 AgSp1.1 glycine rm2 78  
Varen 6869 AgSp1.1 glycine rm1 11  
Mlaby 7048 AgSp1 glycine rm2b 34  
Varen 6869 AgSp1.2 glycine rm1 10  
Msag 18528 AgSp1.2 glycine rm 112  
Mhut 1271 AgSp1 glycine rm1 8  
Varen 6869 AgSp1.2 glycine rm1 14  
Mlaby 7048 AgSp1 glycine rm2b 20  
Msag 18528 AgSp1.2 glycine rm 114  
Aarg AgSp1 FC511 glycine rm1 13  
Aarg AgSp1 FC511 glycine rm2 6  
Msag 18528 AgSp1.2 glycine rm 113  
Varen 6869 AgSp1.2 glycine rm1 1  
Varen 6869 AgSp1.2 glycine rm1 2  
Varen 6869 AgSp1.2 glycine rm1 13  
Varen 6869 AgSp1.2 glycine rm1 11  
Varen 6869 AgSp1.2 glycine rm1 12  
Aarg AgSp1 FC511 glycine rm1 1  
Lcorn 18104 AgSp1.1 glycine rm1 38  
Amarm 50219 AgSp1.1 glycine rm2 1  
Lcorn 18104 AgSp1.1 glycine rm1 37  
Mlaby 7048 AgSp1 glycine rm1a 4  
Aarg AgSp1 FC511 glycine rm1 2  
Mlaby 7048 AgSp1 glycine rm2b 20.1  
consensus

PGTTPGTyTGpDGkpT.KFIVP.GAFITPGSIPGPDGPIH.VEPAGPGTTPGAEtG.PDGKINKIYVPTTPKGPSSP  
PGTTPGTVTGSdGKPK.KFIVPK.PGTTTPGAETG.PDGNVDKIILPTTPVRPEPQ  
PGTTPGYVTDPDGQAV.KFIVPEDAFITPGYIPGPNKGPVR.VGPAG  
PGTTPGVVVTGPDGQPV.QFIVPQGAFTTPGSIPGPNKRIH.VKPAGP  
PGTTPGTVTGPdGQPV.KFIVPQGAFTTPGTILGPDGKPIP.VEPQG  
PGTTPGTVTGPdGKPT.KFIVPKGAFTTPGSIPGPDGKPIH.VEPAG  
PGTTPGTVTGPdGKPK.KFIVVPQGAFTTPGSIPGPDGNPIP.VEPAGPGTTPGTQTG.PDGKINK  
PGTTPGVVVTGPDGQPV.QFIVPQGAFTTPGSIPGPDGKPVH.VRPAGPGTTPGAKNPLPVTQSDGQPIQIV.PA.GP  
PGTTPGTVTGRdGKPR.KFIVPNGAFTTPGSFPGPDGEPIS.VEPAGPGTTPGTIETD.SDGKIKKVVVPTPKGPGEPP  
PGTTPGTVTGPdGKPI.KFIIIPNGAFRTPGSVPGPDGEPIS.VEPAGPGTTPGTETG.PDGKIRKVVVPTPKGPHGP  
PGTTPGTVTGPdGKPS.KFIVPTGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTG.PDGKINKLVLPPTPKAPQGP  
PGTTPGTVTGPdGKPI.QFIVPLGAYTTPGSIPGPDGKPIH.VQAPGPGTTPGTITD.SQGNVDKIILPTTPKGQ  
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PGTTPGTITGPDGRPT.KFIVPYGAFTTPGTIPGPDGKPIQ.VEPAGPGTTPGVERG.DDGKVYKIYLPMLHISPGEL  
PGTTPGTVTGPdGKPI.KFIVPTGAFTTPGSIPGPDGRPIH.VEPAGPGTTPGAETG.PDGKINKIYLPSTPKGPGGP  
PGTTPGTITGPDGKPT.KFIVPYGAFTTPGSIPGPDGRPIH.VEPAGPGTTPGAETG.PDGKINKIYLPSTPKGPGGP  
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PGTTPGTITGSDGRPI.KFIIIPYGAFSTPGSIPGPDGTPIH.VEPAGPGTTPGVKGT.PDGKVYKIYLPSTPKGPGGP  
PGTTPGTITGSDGRPI.QFIIIPYGAYSTPGSIPGPDGTPIH.VEPAGPGTTPGTIETG.PDGKVYKIYLPSTPKGPGGP  
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PGTTPGTETGTdGKPN.KFIVVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAETG.PDGKINKIIVVPTTTKAP  
GTTGTVTGPdGKPT.KFIVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAETG.PDGKINKLVVPTTPKGP  
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PGTTPGTVTGPdGKPK.KFVVPKGAFTTPGSIPGPDGKPIH.VEPAGPGTTPGAQTG.PDGKINKLVVPTTTTPK  
PGSTPGTVTGPdGMPK.KFVVPKGAFTTPGSIPGPDGKPIH.VQAPGPGTTPGAMTG.PDGKINKLVVPTTT.P

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logo

|                                    |                                                                                 |     |
|------------------------------------|---------------------------------------------------------------------------------|-----|
| Amarm 50219 AgSp1.1 glycine rm1 3  | ..GPQGPGGYPLGPGGQ.....T.....                                                    | 10  |
| Msag 18528 AgSp1.2 glycine rm 16   | .....                                                                           | 16  |
| Aarg AgSp1 FC511 glycine rm1 3     | .....                                                                           | 22  |
| Mraby 7048 AgSp1 glycine rm1a 1    | .....PQ.....TTTTTKVKGPQGGKPLVVVPAG.....                                         | 52  |
| Mhut 1271 AgSp1 glycine rm3 17     | .....                                                                           | 45  |
| Mraby 7048 AgSp1 glycine rm1a 6    | .....                                                                           | 46  |
| Aarg AgSp1 FC511 glycine rm2 7     | .....                                                                           | 45  |
| Lcorn 18104 AgSp1.1 glycine rm1 39 | .....                                                                           | 45  |
| Aarg AgSp1 FC511 glycine rm1 15    | .....                                                                           | 62  |
| Mraby 7048 AgSp1 glycine rm1a 15   | .....GTT.....PGVVTGPGGEPLRQVHVR.....PAGPGTTTPGAKNPLPVTSQGQPIQIVPAGPGTTTPGVVTGPG | 136 |
| Varen 6869 AgSp1.1 glycine rm1 2   | .....GGY.....PMRPR.....                                                         | 83  |
| Varen 6869 AgSp1.1 glycine rm1 1   | .....GKM.....FG.PMRPRRQ.....RTTPPSVEGPNKGPIQIEPAG.....                          | 92  |
| Lcorn 18104 AgSp1.1 glycine rm1 1  | .....GIN.....PYYPMRPGR.....                                                     | 87  |
| Mraby 7048 AgSp1 glycine rm1a 2    | .....QFY.....FPV.....TPKPTPTPVTTGPGGGKPIQIIPAG.....                             | 101 |
| Mhut 1271 AgSp1 glycine rm1 1      | .....RTK.....RPSY.THVKDKPIQVIPAG.....                                           | 96  |
| Amarm 50219 AgSp1.1 glycine rm1 2  | .....GG.....QTTTTPVPGR.....                                                     | 88  |
| Mhut 1271 AgSp1 glycine rm1 2      | .....I.....LGKASHGRPIQIIPAG.....                                                | 92  |
| Amarm 50219 AgSp1.1 glycine rm1 8  | .....GFN.....FATP.....ATTPTPIPGPGGGKPIQVIPAG.....                               | 103 |
| Amarm 50219 AgSp1.1 glycine rm1 1  | .....GG.....QTTTTPVPGRGKPIQIIPAG.....                                           | 98  |
| Mhut 1271 AgSp1 glycine rm1 3      | .....GFY.....FG.....MPQYIPGPGGRPIQIVPAG.....                                    | 99  |
| Mhut 1271 AgSp1 glycine rm1 5      | .....GFY.....FG.....MPQYIPGPGGGKPIQIEPAG.....                                   | 96  |
| Mhut 1271 AgSp1 glycine rm1 7      | .....GFY.....FG.....MPQNIIPGPDGKPIQIVPAG.....                                   | 99  |
| Mhut 1271 AgSp1 glycine rm1 4      | .....GFY.....FG.....MPQYIPGPGGRPIQIVPAG.....                                    | 99  |
| Mhut 1271 AgSp1 glycine rm1 6      | .....GFY.....FG.....MPQYIPGPGGGKPIQIEPAG.....                                   | 99  |
| Varen 6869 AgSp1.2 glycine rm1 3   | .....                                                                           | 11  |
| Mraby 7048 AgSp1 glycine rm2b 21   | .....                                                                           | 17  |
| Msag 18528 AgSp1.2 glycine rm 1    | .....                                                                           | 16  |
| Msag 18528 AgSp1.2 glycine rm 115  | .....                                                                           | 35  |
| Amarm 50219 AgSp1.1 glycine rm2 89 | ..APTGPGGFPL.....PGPGGQPIYPYGPSP.....YGPQGGPGG..K.....                          | 35  |
| Amarm 50219 AgSp1.1 glycine rm2 78 | ..APTGPGGFPLRT.....TGPGGQPIYPYGPSP.....YGPQGGPGG..K.....                        | 37  |
| Varen 6869 AgSp1.1 glycine rm1 11  | .....PGGY.....PMGPGN.....PFRPMGPGR..QRTTPSSVRGPKGGKPIQIEPAG.....                | 45  |
| Mraby 7048 AgSp1 glycine rm2b 34   | .....                                                                           | 23  |
| Varen 6869 AgSp1.2 glycine rm1 10  | .....PTIHMGSDSEQGM..NFVTTTPIPGPGGRPIQIEPAG.....                                 | 37  |
| Msag 18528 AgSp1.2 glycine rm 112  | .RGPQGGPGG.....YPLGPGSP.....FGPKGGPGG..QTTTPASVQGGPDGEPIQVIEPAG.....            | 46  |
| Mhut 1271 AgSp1 glycine rm1 8      | .....                                                                           | 45  |
| Varen 6869 AgSp1.2 glycine rm1 14  | .....                                                                           | 45  |
| Mraby 7048 AgSp1 glycine rm2b 20   | ..GPTGPGGQPLN.....PLGPRGQPLNPLGPGSP.....GG..QTTTPAPVPGPDGKPLQIVPAG.....         | 60  |
| Msag 18528 AgSp1.2 glycine rm 114  | .....                                                                           | 70  |
| Aarg AgSp1 FC511 glycine rm1 13    | ..G.....PVGPDGQPMYPSGP.....Q...GPGG..Q.TTTTTPIPGPDGKPIQIEPAG.....               | 91  |
| Aarg AgSp1 FC511 glycine rm2 6     | .....QGPGGY.....PLGPGSP.....FGPKGGPGG..QTTTPASVQGGPDGEPIQVIEPAG.....            | 96  |
| Msag 18528 AgSp1.2 glycine rm 113  | .....                                                                           | 115 |
| Varen 6869 AgSp1.2 glycine rm1 1   | .....PETHPAGRPIQIVPAG.....                                                      | 86  |
| Varen 6869 AgSp1.2 glycine rm1 2   | .....GMN..F.ATTTPIPGPGGRPIQIEPAG.....                                           | 99  |
| Varen 6869 AgSp1.2 glycine rm1 13  | .....GIN..F.ATTTPIPGPGGRPIQIEPAG.....                                           | 99  |
| Varen 6869 AgSp1.2 glycine rm1 11  | .....GMN..F.ATTTPIPGPGGRPIQIEPAG.....                                           | 99  |
| Varen 6869 AgSp1.2 glycine rm1 12  | .....GMN..F.ATTTPIPGPGGRPIQIEPAG.....                                           | 99  |
| Aarg AgSp1 FC511 glycine rm1 1     | .....PEEPSGRPMYFPGPGSP.....YG...PGE..Q.TTTTTPIPGPDGKPLQIEPAG.....               | 114 |
| Lcorn 18104 AgSp1.1 glycine rm1 38 | ....QSGSQ.....PL.....FP.....FGPQGPVG..Q.TTTTTPIPGPDGKPIQIVPAG.....              | 110 |
| Amarm 50219 AgSp1.1 glycine rm2 1  | ....TGPGGFPLGPAG..FPLGPGGQPIYPYGPSP.....YGPQGGPGG..K.TTTTTPIPGPDGEPLSIVPAG..... | 131 |
| Lcorn 18104 AgSp1.1 glycine rm1 37 | ....QGPGLNP....Y..YPMGPEGKPMYPYGPSP.....FGPQGGPGG..Q.TTKTPVPGPDGKPIQIVPAG.....  | 126 |
| Mraby 7048 AgSp1 glycine rm1a 4    | ....TGPGGQPLNPQGPISPFPGGGQSMNPLGPGGS.....GG....Q..T.TTEPIPGNPGKPLQIVPAG.....    | 128 |
| Aarg AgSp1 FC511 glycine rm1 2     | ..GPVGGGMPLSP....YYPQGGGQPMNPFAPGSP.....YG...PGE..Q.TTTTTPIPGPDGKPLQIEPAG.....  | 129 |
| Mraby 7048 AgSp1 glycine rm2b 20.1 | ..GPTGPGGQPLN.....PLGPGGQPLNPLGPGSP.....GG....Q..T.TTPAPVPGPDGKPLQIVPAG.....    | 122 |
| consensus                          | .....                                                                           |     |

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|------------------------------------|----------------------|-----|
| Amarm 50219 AgSp1.1 glycine rm1 3  | .....                | 10  |
| Msag 18528 AgSp1.2 glycine rm 16   | .....                | 16  |
| Aarg AgSp1 FC511 glycine rm1 3     | .....                | 22  |
| Mraby 7048 AgSp1 glycine rm1a 1    | .....                | 52  |
| Mhut 1271 AgSp1 glycine rm3 17     | .....                | 45  |
| Mraby 7048 AgSp1 glycine rm1a 6    | .....                | 46  |
| Aarg AgSp1 FC511 glycine rm2 7     | .....                | 45  |
| Lcorn 18104 AgSp1.1 glycine rm1 39 | .....                | 45  |
| Aarg AgSp1 FC511 glycine rm1 15    | .....                | 62  |
| Mraby 7048 AgSp1 glycine rm1a 15   | GKPLRTTPGPGLQFPTGPGK | 156 |
| Varen 6869 AgSp1.1 glycine rm1 2   | .....                | 83  |
| Varen 6869 AgSp1.1 glycine rm1 1   | .....                | 92  |
| Lcorn 18104 AgSp1.1 glycine rm1 1  | .....                | 87  |
| Mraby 7048 AgSp1 glycine rm1a 2    | .....                | 101 |
| Mhut 1271 AgSp1 glycine rm1 1      | .....                | 96  |
| Amarm 50219 AgSp1.1 glycine rm1 2  | .....                | 88  |
| Mhut 1271 AgSp1 glycine rm1 2      | .....                | 92  |
| Amarm 50219 AgSp1.1 glycine rm1 8  | .....                | 103 |
| Amarm 50219 AgSp1.1 glycine rm1 1  | .....                | 98  |
| Mhut 1271 AgSp1 glycine rm1 3      | .....                | 99  |
| Mhut 1271 AgSp1 glycine rm1 5      | .....                | 96  |
| Mhut 1271 AgSp1 glycine rm1 7      | .....                | 99  |
| Mhut 1271 AgSp1 glycine rm1 4      | .....                | 99  |
| Mhut 1271 AgSp1 glycine rm1 6      | .....                | 99  |
| Varen 6869 AgSp1.2 glycine rm1 3   | .....                | 11  |
| Mraby 7048 AgSp1 glycine rm2b 21   | .....                | 17  |
| Msag 18528 AgSp1.2 glycine rm 1    | .....                | 16  |
| Msag 18528 AgSp1.2 glycine rm 115  | .....                | 35  |
| Amarm 50219 AgSp1.1 glycine rm2 89 | .....                | 35  |
| Amarm 50219 AgSp1.1 glycine rm2 78 | .....                | 37  |
| Varen 6869 AgSp1.1 glycine rm1 11  | .....                | 45  |
| Mraby 7048 AgSp1 glycine rm2b 34   | .....                | 23  |
| Varen 6869 AgSp1.2 glycine rm1 10  | .....                | 37  |
| Msag 18528 AgSp1.2 glycine rm 112  | .....                | 46  |
| Mhut 1271 AgSp1 glycine rm1 8      | .....                | 45  |
| Varen 6869 AgSp1.2 glycine rm1 14  | .....                | 45  |
| Mraby 7048 AgSp1 glycine rm2b 20   | .....                | 60  |
| Msag 18528 AgSp1.2 glycine rm 114  | .....                | 70  |
| Aarg AgSp1 FC511 glycine rm1 13    | .....                | 91  |
| Aarg AgSp1 FC511 glycine rm2 6     | .....                | 96  |
| Msag 18528 AgSp1.2 glycine rm 113  | .....                | 115 |
| Varen 6869 AgSp1.2 glycine rm1 1   | .....                | 86  |
| Varen 6869 AgSp1.2 glycine rm1 2   | .....                | 99  |
| Varen 6869 AgSp1.2 glycine rm1 13  | .....                | 99  |
| Varen 6869 AgSp1.2 glycine rm1 11  | .....                | 99  |
| Varen 6869 AgSp1.2 glycine rm1 12  | .....                | 99  |
| Aarg AgSp1 FC511 glycine rm1 1     | .....                | 114 |
| Lcorn 18104 AgSp1.1 glycine rm1 38 | .....                | 110 |
| Amarm 50219 AgSp1.1 glycine rm2 1  | .....                | 131 |
| Lcorn 18104 AgSp1.1 glycine rm1 37 | .....                | 126 |
| Mraby 7048 AgSp1 glycine rm1a 4    | .....                | 128 |
| Aarg AgSp1 FC511 glycine rm1 2     | .....                | 129 |
| Mraby 7048 AgSp1 glycine rm2b 20.1 | .....                | 122 |
| consensus                          |                      |     |

 non-conserved  
 similar  
  $\geq 50\%$  conserved